

No.	Solution
1(a)	As the cyclist pedals forward, the ground exerts a forward force on the bicycle that is equal and opposite to the frictional forces opposing the motion of the cyclist. Thus there is no net force acting on the cyclist, and the cyclist will continue in his uniform state of motion at constant speed in a straight line according to Newton's first law of motion.
1(b)	Making v the subject, $v = \sqrt{\frac{2F}{c_D \rho A}}$ $= \sqrt{\frac{2(22)}{(0.88)(1.2)(0.32)}}$ $= 11.4 \text{ m s}^{-1}$ $\frac{\Delta v}{v} = 0.5 \left(\frac{\Delta F}{F} + \frac{\Delta c_D}{c_D} + \frac{\Delta \rho}{\rho} + \frac{\Delta A}{A} \right)$ $= 0.5 \left(\frac{2}{22} + \frac{0.01}{0.88} + \frac{0.1}{1.2} + \frac{0.02}{0.32} \right)$ $= 0.124$ $\Delta v = 0.124(11.41)$ $= 1.41$ $\approx 1 \text{ m s}^{-1} \text{ (1 s.f.)}$ Hence, $v = (11 \pm 1) \text{ m s}^{-1}$
1(c)(i)	Let the driving force be F_D and the displacement in the direction of the force be s. Work done, $W = F_D \times s$ Since net force = 0, $F_D = \text{drag force} = 22 \text{ N}$ power output = rate of work done $= \frac{dW}{dt} = \frac{d(22s)}{dt} = 22 \frac{ds}{dt} = 22v$
1(c)(ii)	power output = $22v = 22 \times 11 = 240 \text{ W}$
2(a)	Gravitational potential at a point is the work done per unit mass in bringing a small